Solution: Week 6 practice assignment.

1) Option 1:

If T is a linear transformation, then

$$T(4,1) = T((2,3) + 2(1,-1))$$

$$= T(2,3) + 2T(1,-1)$$

$$= (1,2) + 2(1,-1) = (3,0)$$

But it is given that $T(4,1) = (1,0) \neq (3,0)$

Hence, the statement is wrong.

Option 2:

$$T(1,0) = T\left(\tfrac{1}{5}(2,3) + \tfrac{3}{5}(1,-1)\right)$$

$$= \tfrac{1}{5}\, T(2,3) + \tfrac{3}{5}\, T(1,-1)$$

$$= \tfrac{1}{5}(1,2) + \tfrac{3}{5}(1,-1)$$

$$= \left(\tfrac{4}{5}, -\tfrac{1}{5}\right) = \tfrac{1}{5}(4,-1)$$

$$T(0,1) = T\left(\tfrac{1}{5}(2,3) - \tfrac{2}{5}(1,-1)\right)$$

$$= \tfrac{1}{5}\, T(2,3) - \tfrac{2}{5}\, T(1,-1)$$

$$= \tfrac{1}{5}(1,2) - \tfrac{2}{5}(1,-1) = \left(-\tfrac{1}{5}, \tfrac{4}{5}\right) = \tfrac{1}{5}(-1,4)$$

$$T(x,y) = T\left(x(1,0) + y(0,1)\right)$$
$$= x\,T(1,0) + y\,T(0,1) = \tfrac{1}{5}(4x,-x) + \tfrac{1}{5}(-y,4y)$$
$$= \tfrac{1}{5}(4x-y,\ -x+4y)$$

Hence, the statement is true.

<u>Option 3</u>:

Basis of $V = \{v_1, v_2, v_3\}$

Basis of $W = \{2v_1 + v_3,\ v_2 - v_3,\ v_3 - v_1\}$

$$T(v_1) = 2v_3 + v_1 = 1(2v_1 + v_3) + 0(v_2 - v_3) + 1(v_3 - v_1)$$

$$T(v_2) = 2v_1 = \tfrac{2}{3}(2v_1 + v_3) + 0(v_2 - v_3) - \tfrac{2}{3}(v_3 - v_1)$$

$$T(v_3) = 2v_2 = \tfrac{2}{3}(2v_1 + v_3) + 2(v_2 - v_3) + \tfrac{4}{3}(v_3 - v_1)$$

Hence, the matrix representation of T with respect to the given bases is, $\begin{pmatrix} 1 & 2/3 & 2/3 \\ 0 & 0 & 2 \\ 1 & -2/3 & 4/3 \end{pmatrix}$

<u>option - 1</u> :

$$\begin{pmatrix} 1 & 2/3 & 2/3 \\ 0 & 0 & 2 \\ 1 & -2/3 & 4/3 \end{pmatrix} \xrightarrow{R_3 - R_1} \begin{pmatrix} 1 & 2/3 & 2/3 \\ 0 & 0 & 2 \\ 0 & -4/3 & 2/3 \end{pmatrix}$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 2/3 & 2/3 \\ 0 & -4/3 & 2/3 \\ 0 & 0 & 2 \end{pmatrix}$$

$$\xrightarrow{-3/4\, R_2} \begin{pmatrix} 1 & 2/3 & 2/3 \\ 0 & 1 & -1/2 \\ 0 & 0 & 2 \end{pmatrix}$$

$$\xrightarrow{R_1 - 2/3 R_2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1/2 \\ 0 & 0 & 2 \end{pmatrix}$$

$$\xrightarrow{\frac{R_3}{2}} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1/2 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_1 - R_3} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1/2 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_2 + 1/2 R_3} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Hence rank of T is 3 .

2) i) $T : \mathbb{R}^2 \to \mathbb{R}^2$

Matrix representation of T with respect to standard ordered basis $\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$

$T(1,0) = (1,-1)$, $T(0,1) = (-1,1)$

$T(x,y) = (x-y, -x+y)$

$$S = \{(x, y) \mid x+y=1,\ x, y \in \mathbb{R}\}$$
$$= \{(x, 1-x) \mid x \in \mathbb{R}\}$$

$$T(x, 1-x) = (2x-1,\ -2x+1)$$
$$= (x', y') \quad \text{(say)}$$

$$x' = 2x-1, \quad y' = -2x+1$$

$$x' + y' = 2x - 1 - 2x + 1 = 0$$

Hence, $T(S) = \{(x, y) \mid x+y=0,\ x, y \in \mathbb{R}\}$

$x+y=0$ is the st. line which passes through origin and has negative slope.

Hence, i) $\longrightarrow$ b) $\longrightarrow$ 3)

ii) $T: \mathbb{R}^2 \to \mathbb{R}^2$

The matrix representation of T with respect to standard ordered basis is $\begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}$

$T(1,0) = (0,1)$ and $T(0,1) = (-1, -1)$

$T(x, y) = (-y, x-y)$

$S = \{(x, y) \mid x + y = 1, x, y \in \mathbb{R}\}$

$= \{(x, 1-x) \mid x \in \mathbb{R}\}$

$T(x, 1-x) = (-1 + x, 2x - 1)$

$= (x', y')$ (say)

$x' = -1 + x, \quad y' = 2x - 1$

$2x' = -2 + 2x, \quad -y' = -2x + 1$

$2x' - y' = -1$

Hence, $T(S) = \{(x, y) \mid 2x - y = -1, x, y \in \mathbb{R}\}$

$2x - y = -1$ is a st. line which has positive slope and whose y-intercept is positive.

Hence, ii) $\rightarrow$ c $\rightarrow$ 2)

iii) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$

The matrix representation of T with respect to standard ordered basis is

$$\begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}$$

$T(1, 0) = (1, -1)$ and $T(0, 1) = (1, 0)$

$T(x, y) = (x+y, -x)$

$S = \{(x, y) \mid x+y=1, \ x, y \in \mathbb{R}\}$

$= \{(x, 1-x) \mid x \in \mathbb{R}\}$

$T(x, 1-x) = (1, -x)$

$T(S) = \{(x, y) \mid x = 1, \ y \in \mathbb{R}\}$

$x = 1$ is the st. line parallel to Y-axis passing through $(1, 0)$.

Hence, iii) $\rightarrow$ a) $\rightarrow$ 1

3) $T : \mathbb{R}^2 \to \mathbb{R}^2$

$$T(x,y) = \begin{cases} (x^2/y, \ y) & , \text{ if } y \neq 0 \\ (x, 0) & , \text{ otherwise} \end{cases}$$

Let $v_1 = (1,1)$ and $v_2 = (1,2)$

$T(1,1) = (1,1)$, $T(1,2) = (1/2, 2)$

$T((1,1)+(1,2)) = T(2,3) = (4/3, 3)$

Hence, $T(v_1 + v_2) \neq T(v_1) + T(v_2)$

If $y \neq 0$, then $cy \neq 0$ for all $c \in \mathbb{R} \setminus \{0\}$.

$$T(cx, cy) = \left(\frac{c^2x^2}{cy}, cy\right)$$

$$= \left(cx^2/y, cy\right)$$

$$= c\left(x^2/y, y\right)$$

And if $y \neq 0$ and $c = 0 \Rightarrow cx = cy = 0$, so $T(0,0) = (0,0) = 0\,T(x,y)$

So $T(cx, cy) = c\,(x,y)$.

If $y = 0$, then

$$T(cx, 0) = (cx, 0) = c(x, 0)$$

Hence, $T(cv) = c\,T(v)$ for all $v \in \mathbb{R}^2$ and $c \in \mathbb{R}$.

Hence, P satisfies 2 but not 1.

$$T : \mathbb{R}^2 \to \mathbb{R}^2$$

$$T(x, y) = (x, xy)$$

Let $v_1 = (1, 2)$, $v_2 = (1, 1)$

$$T(v_1 + v_2) = T(2, 3) = (2, 6)$$

$$T(v_1) + T(v_2) = T(1, 2) + T(1, 1)$$

$$= (1, 2) + (1, 1) = (2, 3)$$

Hence, 1 is not satisfied.

Let $v = (1,2)$ and $c = 2$

Then $T(cv) = T(2,4) = (2,8)$

$cT(v) = 2(1,2) = (2,4)$

Hence, $T(cv) \neq cT(v)$

Hence θ doesnot satisfy 1, 2 and 3.

$T: \mathbb{R}^3 \to \mathbb{R}$

$$T(x,y,z) = \frac{x+y+z}{5}$$

Let $v_1 = (x_1, y_1, z_1)$ and $v_2 = (x_2, y_2, z_2)$

$$T(v_1 + v_2) = T(x_1 + x_2, y_1 + y_2, z_1 + z_2)$$

$$= \frac{x_1 + x_2 + y_1 + y_2 + z_1 + z_2}{5}$$

$$= \frac{x_1 + y_1 + z_1}{5} + \frac{x_2 + y_2 + z_2}{5}$$

$$= T(x_1, y_1, z_1) + T(x_2, y_2, z_2)$$

$$= T(v_1) + T(v_2)$$

4) $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$

$\beta = \{(1,0,0), (1,1,0), (1,1,1)\}$

$\gamma = \{(1,1,1), (0,1,1), (0,0,1)\}$

The matrix representation in (with respect to β and γ) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$T(1,0,0) = 1(1,1,1) + 0(0,1,1) + 0(0,0,1) = (1,1,1)$
$T(1,1,0) = 0(1,1,1) + 1(0,1,1) + 0(0,0,1) = (0,1,1)$
$T(1,1,1) = 0(1,1,1) + 0(0,1,1) + 1(0,0,1) = (0,0,1)$

Standard ordered basis : $\{(1,0,0), (0,1,0), (0,0,1)\}$

$(1,0,0) = 1(1,0,0) + 0(1,1,0) + 0(1,1,1)$
$(0,1,0) = -1(1,0,0) + 1(1,1,0) + 0(1,1,1)$
$(0,0,1) = 0(1,0,0) - 1(1,1,0) + 1(1,1,1)$

$T(1,0,0) = 1\,T(1,0,0) = (1,1,1)$
$T(0,1,0) = -1\,T(1,0,0) + 1\,T(1,1,0) = -(1,1,1) + (0,1,1)$
$= (-1,0,0)$

$T(0,0,1) = -1\,T(1,1,0) + 1\,T(1,1,1) = -1(0,1,1) + (0,0,1)$
$= (0,-1,0)$

Hence the matrix representation of T with respect to standard ordered basis is

$$A = \begin{pmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 1 & 0 & 0 \end{pmatrix}$$

Now, need to find $\det(A)$.

So $\det(A) = 1 \det\begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix} + 1 \det\begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix} + 0$

$\det(A) = 1$

5) $(S+T) : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$

$$(S+T)(x,y,z) = S(x,y,z) + T(x,y,z)$$

$$= (x+y,\ y+z,\ z+x) + (x-y,\ y-z,\ z-x)$$

$$= (2x,\ 2y,\ 2z)$$

$$(S+T)(1,0,0) = (2,0,0)$$
$$(S+T)(0,1,0) = (0,2,0)$$
$$(S+T)(0,0,1) = (0,0,2)$$

Hence, $C = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} = 2\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$$= 2I.$$

So, $n = 2$.

Question 6: Consider a linear transformation $T: \mathbb{R}^3 \to \mathbb{R}^3$ defined as $T(x,y,z) = (x+y, z-x, y+z)$.

Option 2: Standard ordered basis of $\mathbb{R}^3$ is $\{(1,0,0),(0,1,0),(0,0,1)\}$.

So $T(1,0,0) = (1,-1,0) = 1(1,0,0) - 1(0,1,0) + 0(0,0,1)$

$T(0,1,0) = (1,0,1) = 1(1,0,0) + 0(0,1,0) + 1(0,0,1)$

$T(0,0,1) = (0,1,1) = 0(1,0,0) + 1(0,1,0) + 1(0,0,1)$

Hence matrix representation of T w.r.t the standard ordered bases is (let) $A = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

Option 3: · Observe that rank of the matrix A (as above) is 3.

Hence, T is a bijective linear transformation.

And $\{(1,0,0),(0,1,0),(0,0,1)\}$ is a linearly independent set.

Hence $\{(1,0,0),(0,1,0),(0,0,1)\}$ is a basis of the range of T.

Option 5: · Observe that rank of $(T) =$ rank (A)
$= 3$

Option 8: Use rank nullity theorem

$$\text{rank}(T) + \text{nullity}(T) = \dim(\mathbb{R}^3)$$

$$\Rightarrow \text{nullity}(T) = 0.$$

Question 7: Let V be a vector space with ordered basis $\alpha = \{v_1, v_2, v_3, v_4\}$. Consider a linear transformation $T(v_1) = v_1$ and $T(v_i) = v_i - v_{i-1}$, where $i > 1$. Let A be the matrix representation of the linear transformation T, with respect to α for both domain & codomain. Then the trace of A is.

Solution: Given $T(v_1) = v_1$ or $T(v_1) = v_1 + 0v_2 + 0v_3 + 0v_4$

and $T(v_i) = v_i - v_{i-1}$

So $T(v_2) = v_2 - v_1 = -v_1 + v_2 + 0v_3 + 0v_4$

$T(v_3) = v_3 - v_2 = 0v_1 - v_2 + v_3 + 0v_4$

$T(v_4) = v_4 - v_3 = 0v_1 + 0v_2 - v_3 + v_4$

Hence $A = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Now trace of (A) = Sum of the diagonal entries

$= 1 + 1 + 1 + 1$

$= 4$

Comprehension Type Question:

$T(3, 10, 8) = 3$

$T(4, 15, 10) = 2$

$T(5, 20, 15) = 1$

8) $3T(5, 20, 15) - T(3, 10, 8) = 3 - 3 = 0$

$\Rightarrow T(15, 60, 45) - T(3, 10, 8) = 0$

$\Rightarrow T(12, 50, 37) = 0$

$$2T(5, 20, 15) - T(4, 15, 10) = 2 - 2 = 0$$

$$\Rightarrow T(10, 40, 30) - T(4, 15, 10) = 0$$

$$\Rightarrow T(6, 25, 20) = 0$$

$$\Rightarrow 2T(6, 25, 20) = 0$$

$$\Rightarrow T(12, 50, 40) = 0$$

Hence we have, $T(12, 50, 40) - T(12, 50, 37) = 0$

$$\Rightarrow T(0, 0, 3) = 0$$

$$\Rightarrow T(0, 0, 1) = 0.$$

Moreover we have, $20T(0, 0, 1) = 0$

$$\Rightarrow T(0, 0, 20) = 0$$

So, $T(6, 25, 20) - T(0, 0, 20) = 0$

$$\Rightarrow T(6, 25, 0) = 0$$

T is a linear transformation from a 3 dimensional vector space to $\mathbb{R}$, which is a one dimensional vector space over $\mathbb{R}$.

As T is non-zero, rank $(T) = 1$

From rank nullity theorem, we get,

$$\text{nullity}(T) = 3 - 1 = 2$$

Now, $\{(6,25,0), (0,0,1)\}$ is a linearly independent set and we have already derived nullity$(T) = 2$.

So, $\{(6,25,0), (0,0,1)\}$ forms a basis of nullity(T).

In the given options the following will be bases of nullity(T):

option2: $\{(6,25,0), 25(0,0,1)\} = \{(6,25,0), (0,0,25)\}$

Option4: $\{2(6,25,0), (0,0,1)\} = \{(12,50,0), (0,0,1)\}$

option5: $\{4(6,25,0), 25(0,0,1)\} = \{(24,100,0), (0,0,25)\}$

g) $2T(3,10,8) - T(5,20,15) = (2\times 3) - 1 = 5$

$\Rightarrow T(6,20,16) - T(5,20,15) = 5$

$\Rightarrow T(1,0,1) = 5$

$\Rightarrow T(1,0,0) + T(0,0,1) = 5$

$\Rightarrow T(1,0,0) + 0 = 5$

$\Rightarrow T(1,0,0) = 5$

$T(5,20,15) = 1$

$\Rightarrow T(5,0,0) + T(0,20,0) + T(0,0,15) = 1$

$\Rightarrow \quad 5T(1,0,0) + 20\,T(0,1,0) + 15\,T(0,0,1) = 1$

$\Rightarrow \quad 25 + 20\,T(0,1,0) + 0 = 1$

$\Rightarrow \quad T(0,1,0) = -24/20 = -6/5$

$T(x,y,z) = 5x - \frac{6y}{5}$

$T(3,0,0) = 15$ and $T(0,0,1) = 0$

Hence the first three options are correct.

$T(9,0,0) + T(0,25,0) + T(0,0,1)$

$= 45 - 30 = 15$

Hence option 5 and 7 are also correct.

$T(25,9,0) = \left(125 - \frac{54}{5}\right)$ not a multiple of 15.

Hence option 6 and 8 are not correct.

$T(0,0,1) = 0$, which is a multiple of 15.

10) $T(x,y,z) = 5x - \frac{6y}{5}$

$y = 5$

$T(x,5,z) = 5x - 6 = 4$

$\Rightarrow \underline{x = 2}$.